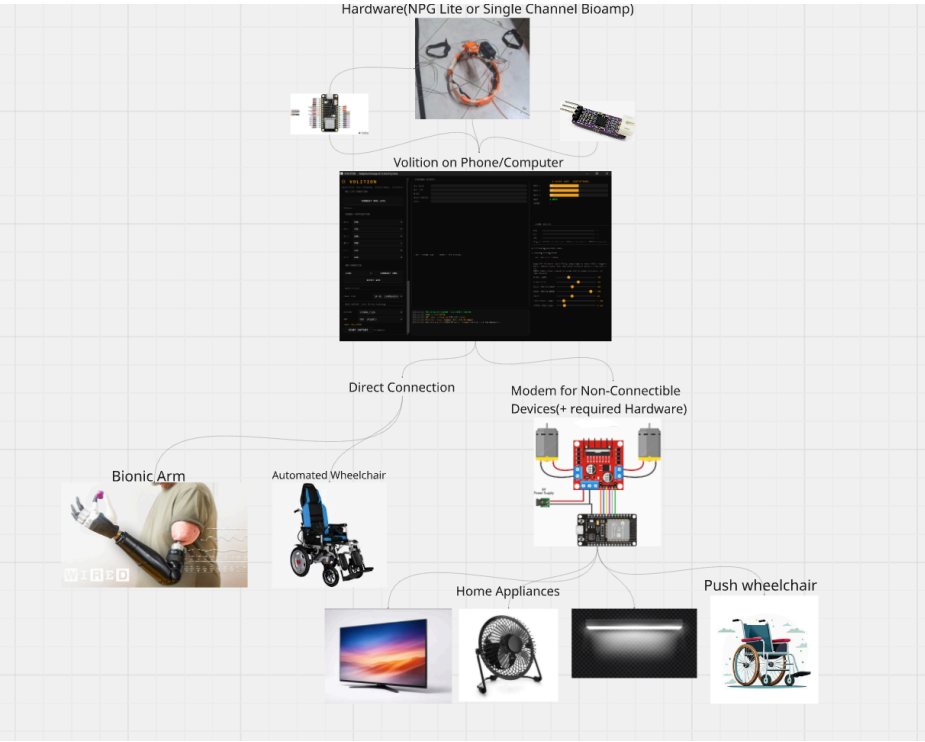
VOLITION

Adaptive Biosignal Platform for Universal Assistive Control

vivo Ignite 4th Edition (2026) · "Tech for Good"
Category: Community Development Innovations / HealthTech Innovations



Document Purpose
Supplementary Technical Reference & Engineering Dossier accompanying the Main Portal Application Submission (Project Display Board).

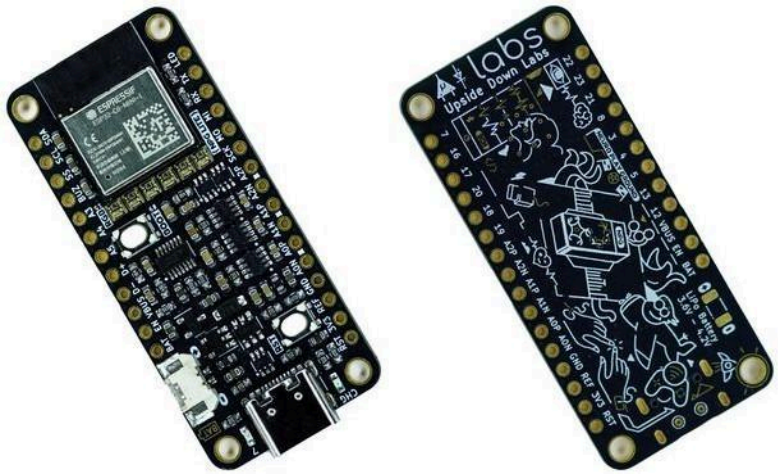
Supporting Technical Document
Hardware reference, system architecture, prototype documentation, cost breakdown and SDG alignment

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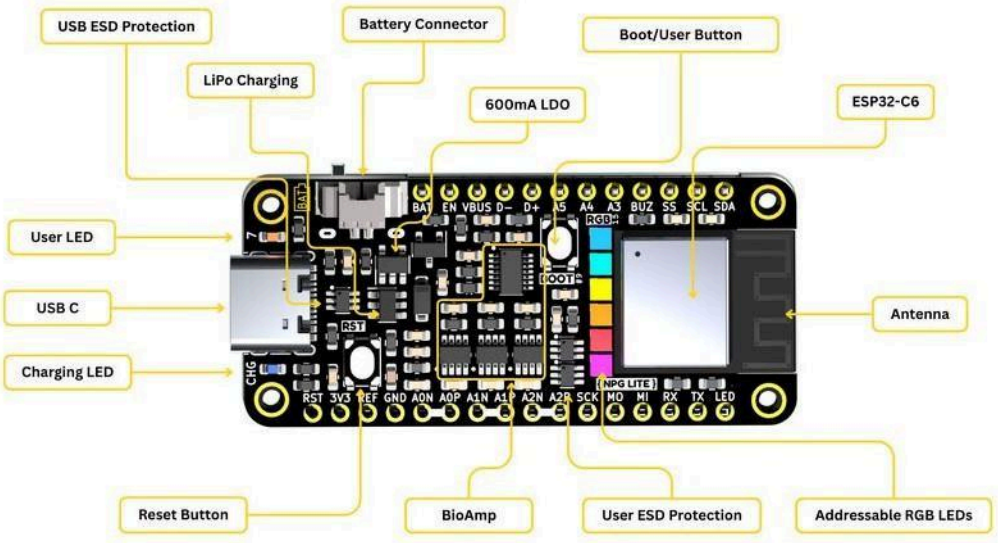
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Hardware & Acquisition

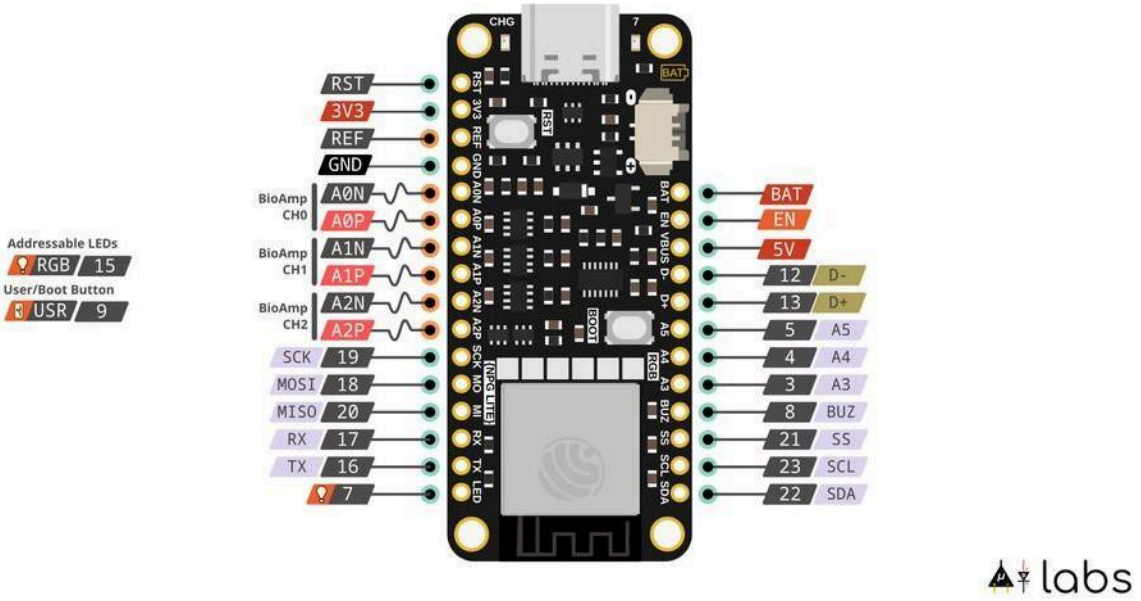
Volition is built on the Neuro PlayGround Lite (NPG Lite), an open-source biosignal acquisition board designed by Upside Down Labs. It integrates an ESP32-C6 microcontroller with Wi-Fi 6 and Bluetooth 5 Low Energy, a 3-channel BioAmp for capturing EMG/EEG/EOG/ECG signals, and onboard LiPo charging. In this project I used it alongside its VIBZ Playmate expansion board.



NPG Lite board — front (BioAmp Chips + ESP32-C6 chip) and back with pin labels.



Onboard Components of the NPG Lite



NPG Lite pin assignment — 3 BioAmp channels (A0/A1/A2) used for variable purposes, but in this project as EEG, EMG and EMG respectively

Component Reference

1	**ESP32-C6**	The ESP32-C6 is a Wi-Fi 6 and Bluetooth 5 (LE) enabled microcontroller developed by Espressif Systems.
2	**Antenna**	The onboard PCB antenna is tuned for 2.4GHz, ensuring strong, stable Wi-Fi and Bluetooth connectivity with low power use.
3	**6 Addressable RGB LEDs**	Enables visualization of device states, signal strength, or neural feedback using dynamic color indicators.
4	**User ESD Protection**	ESD protection components, ensuring durability and reliability when connecting electrodes.
5	**BioAmp**	The onboard BioAmp is a low-noise, high-gain biopotential amplifier that captures and amplifies EMG, ECG, and EEG signals for digital processing by the ESP32-C6.
6	**Boot/User Button**	The Boot/User button enables bootloader mode for firmware flashing and serves as a programmable user input.
7	**600mA LDO**	The voltage regulator converts higher input voltages (USB or battery) into a stable 3.3V output.
8	**Battery Connector**	It offers a safe, port for battery power, allowing portable operation of the board. **Note:** There is no polarity protection.
9	**LiPo Charging**	The integrated LiPo charging controller safely manages charging cycles, supporting automatic termination and trickle charging for optimal battery health.
10	**USB ESD Protection**	USB ESD protection shields the interface from electrostatic and voltage spikes, ensuring reliable data transfer and preventing component damage.
11	**Reset Button**	The reset button restarts the microcontroller and running program without power loss, aiding debugging and firmware testing.
12	**Charging LED**	The Charging LED indicates battery charging status, lighting during charging and turning off when fully charged for immediate feedback.
13	**USB C**	USB C port for data transfer and power input.
14	**User LED**	A programmable LED on a GPIO pin used to show board activity, connection status, or custom signals during development.

Figure 4: Onboard component reference table

The Workings of Volition & System Architecture

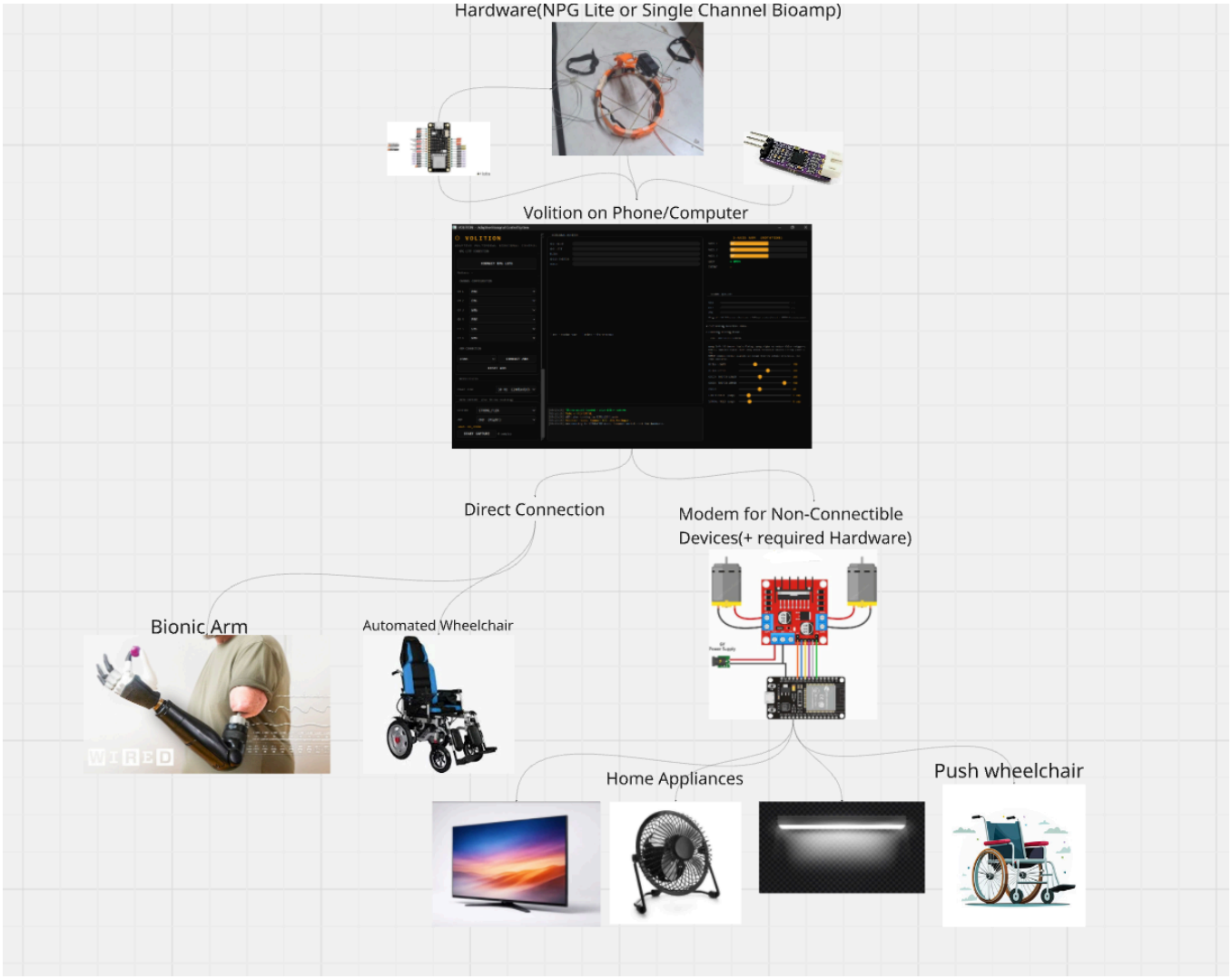


Figure: Illustration showing the map of working of Volition

Volition is an easy to use, ML - aided application that connects to a large variety of devices.

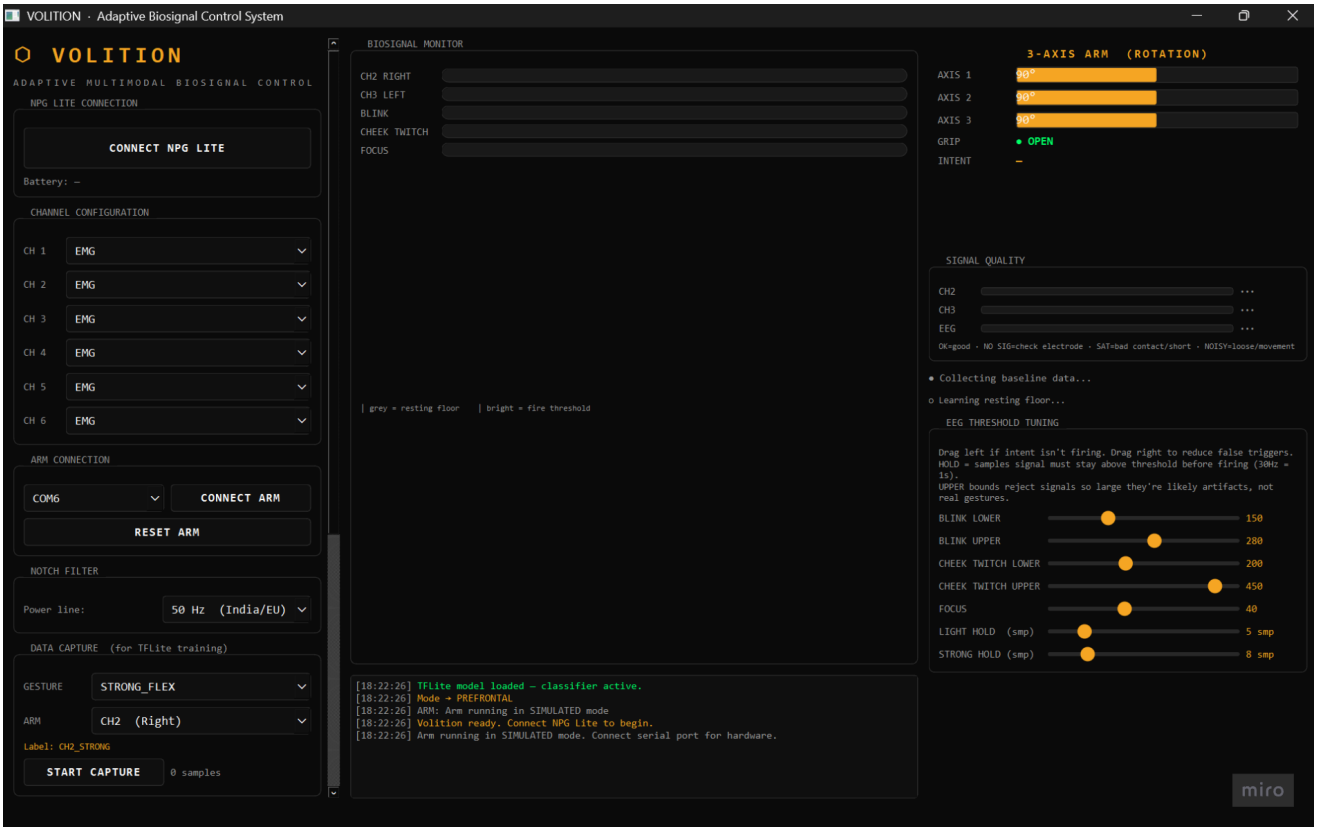


Figure: The Main Volition Application

Signals are acquired from three BioAmp channels, streamed over Bluetooth Low Energy, and processed through a layered software pipeline before reaching the robotic arm or a connected computer.

Inputs and parts of Volition

Stage	Function
NPG Lite (ESP32-C6)	Acquires raw EMG/EEG at 500Hz, streams via BLE
Channel Processor	Per-channel bandpass + notch (50/60Hz) filtering, and signal envelope detection
Passive Floor Tracker	Continuously learns each channel's resting baseline in the background
Signal Health Monitor	Detects disconnected electrodes, saturation, and noise from raw ADC data
Intent Engine	Adaptive thresholds + on-device TFLite model to classify general gestures
Controller / HID	Dispatches classified intents to interfaced devices and computer applications

Different biopotential signals differ by electrode placement, so Volition supports two selectable modes, each with its own calibration profile and on-device model.

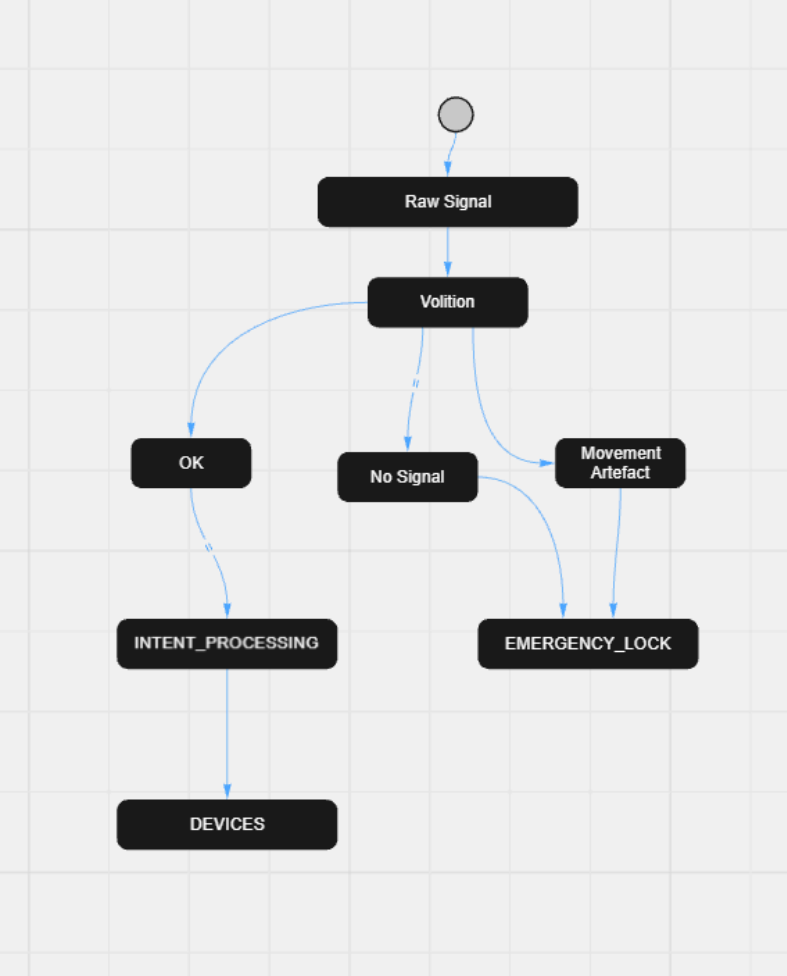


Figure: A flowchart showing the streamlined process of detection on Volition.

Two Headband Detection Modes

Mode	Placement	Primary Signals
Prefrontal (PFC)	Forehead (Fp1/Fp2)	Cheek twitch, jaw clench, focus (blink optional, reduced amplitu
Visual Cortex (VC)	Back of head (O1/O2)	Single / double / triple blink, focus

Each headband placement mode has it's own characteristic biopotential signal that cannot be easily detected in the other. For example, the Prefrontal cortex mode can detect cheek twitches but not blinks whereas the Visual Cortex mode can easily detect blinks but not cheek twitches

Physical Prototype

I employed the use of a custom headset (that I created) for Volition.

The headset houses three dry electrodes and 4 electrode slots (currently frontal positive and negative, with one lateral reference electrode) and mounts the NPG Lite directly on the band.

I designed the headset to be adjustable (with the mechanism on the back) and to maintain electrode positioning for as long as possible. The support bracket on the top was molded to the shape of my head using a hairdryer.



Figure: Volition headset prototype, NPG Lite mounted on the adjustable band with dry electrode contacts.



Figure: Another angle of image of the Volition Interface. The supportive bracket holding the NPG Lite is visible on the top.

Future Scalability and Expansion

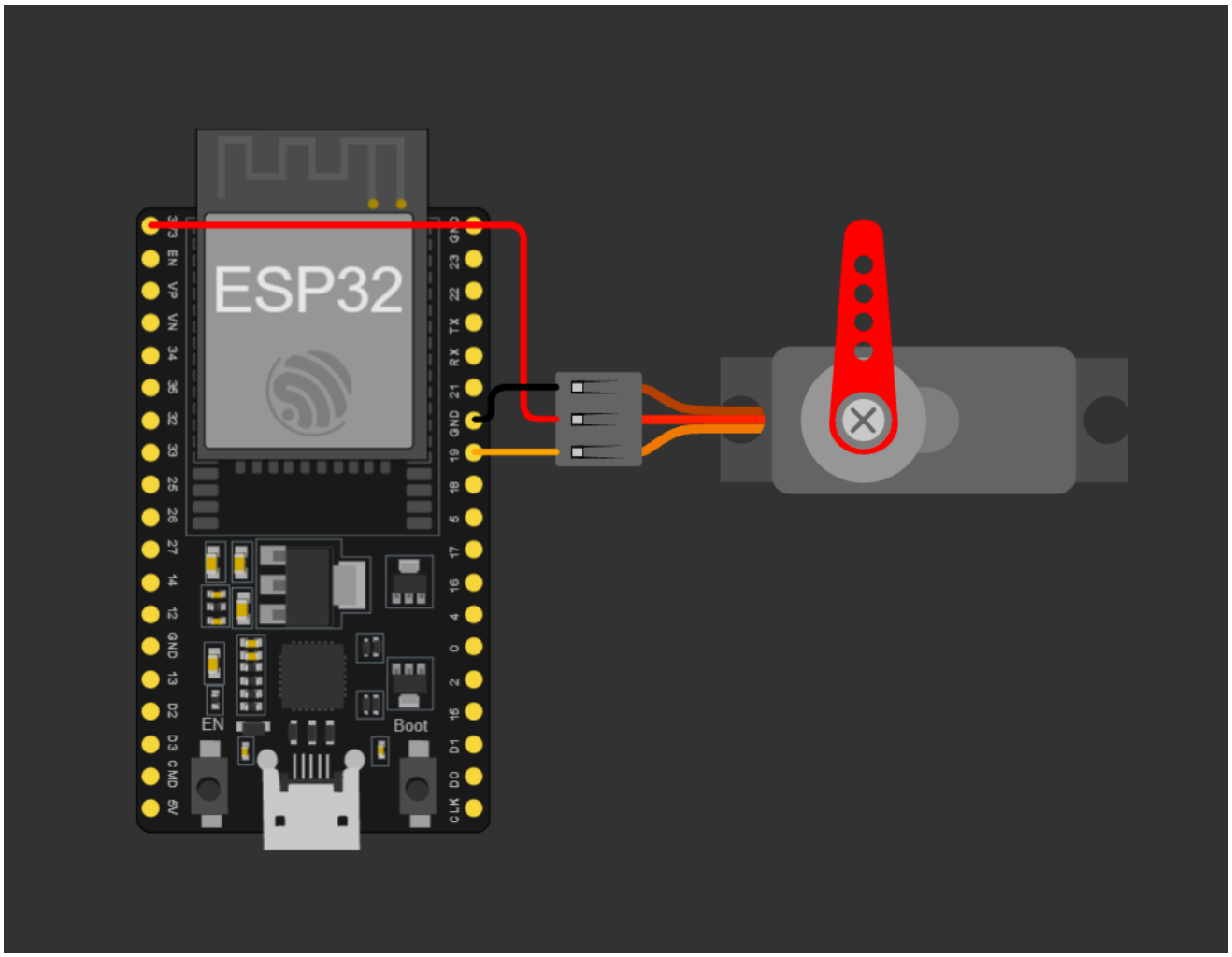


Figure: Schematic blueprint for optional controllable relay and motor driver expansions to interface with physical components.

While the core Volition application achieves out-of-the-box utility through direct wireless connection to computers, smart devices, and robotic aids, its architecture is engineered for long-term modular scalability.

For advanced or high-power use cases, the control platform can be expanded using standard external hardware interfaces. These expansions can be scaled dynamically based on the complexity of the machinery:

- Basic Appliance Control: A simple external servo motor setup powered by an ESP32 to mechanically interact with standard wall switches.
- Heavy Mobility Automation: A modular upgrade path integrating high-capacity battery packs, heavy-duty motor drivers, and wiper motors to automate traditional push wheelchairs.

Cost Breakdown (commercial viability)

Estimated cost using the open-source NPG Lite schematics built as a custom board, versus commercial retail hardware carrying the same functionality:

Component	Est. Cost (INR)
Custom PCB (NPG Lite open schematic)	~2,500
ESP32-C6 module + BioAmp components	included above
Reusable dry electrodes (x3)	~200
Headband materials (printed/reused)	~150
Misc. (wiring, connectors, enclosure)	~500-800
Additional Components for Future Expansion	~800 - 4,500
Approximate Total	**~4,000-8,500**

For comparison, commercial myoelectric prosthetic arms and smart wheelchairs in India retail between Rs. 1-8 lakh, typically requiring professional clinical calibration and, for several models, an internet connection for configuration and updates.

UN Sustainable Development Goal Alignment

SDG	Alignment
SDG 3 — Good Health and Well-Being	Restores physical interaction ability for people with severe motor impairments, directly improving quality of life and independence.
SDG 10 — Reduced Inequalities	Closes the gap between commercial assistive technology (Rs. 1-8 lakh) and what a low-income Indian family can access (~Rs. 4,200), removing income as a barrier to autonomy.
SDG 9 — Industry, Innovation and Infrastructure	Built on open-source Indian hardware (Upside Down Labs) with offline on-device machine learning, creating a reproducible blueprint for local manufacture and adoption.
SDG 17 — Partnerships for the Goals	Relies entirely on open-source collaboration — UDL's public schematics, TensorFlow Lite, and openly documented BLE protocols.

This document is a supplementary technical reference prepared to accompany the Project Display Board submission for vivo Ignite 4th Edition (2026). Hardware images and schematics are the property of Upside Down Labs and are reproduced here for reference with attribution.

Why Volition Might Fail

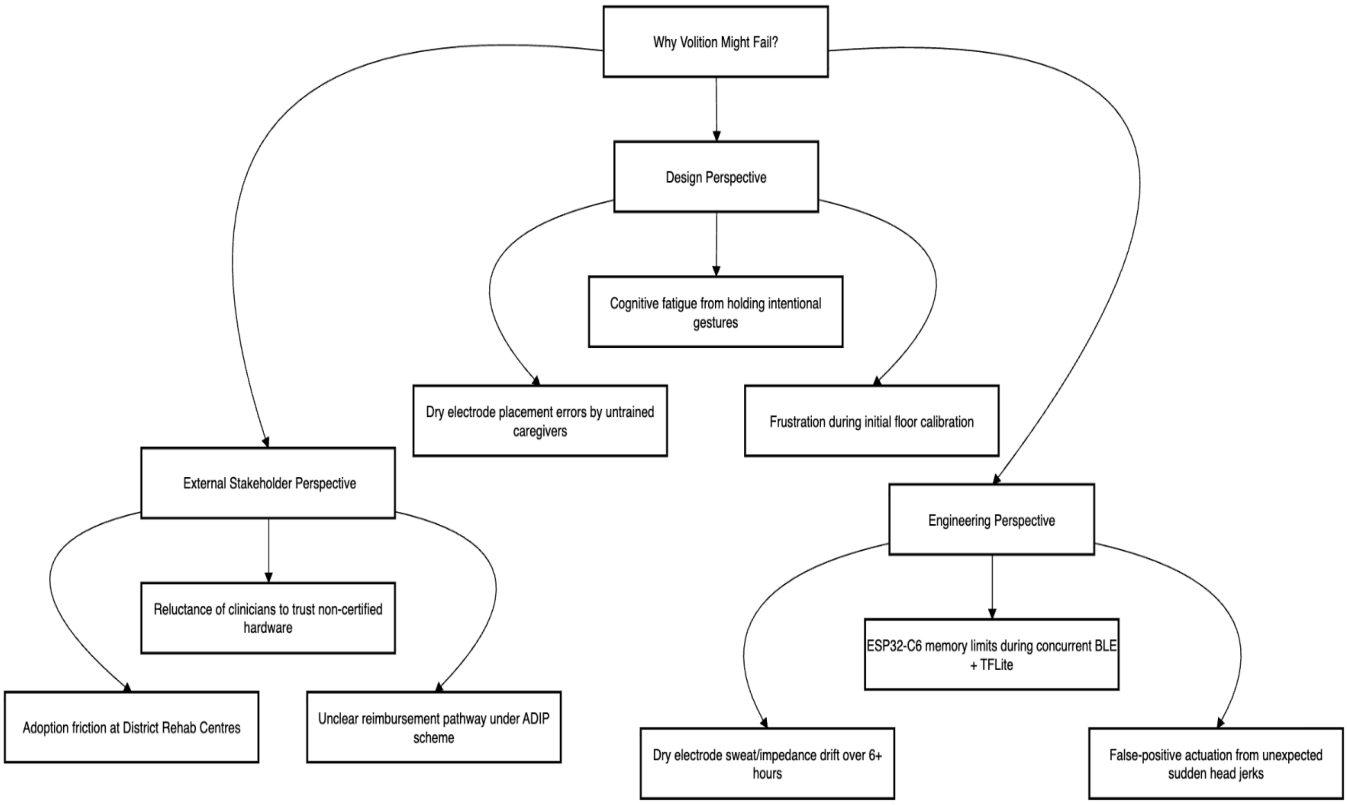


Figure: why Volition might fail, due to multiple purposes like user reluctance and hardware failure.

Although Volition is designed to be an extremely usable product, there are still a multitude of reasons as to why it might fail.

One of the most likely reasons is because the hardware itself may record errors due to improper placement of electrodes, and although we can detect errors in placement the user unfortunately may not be able to mitigate it by themselves and may rely on their caregivers.

Another likely reason is due to caregiver limitations and mindset. Volition's interface would be a foreign item in rural and underdeveloped areas, and may not be utilized simply because of the oddness of wearing it.